

EXP – 1.2

Program:

```
clc;
clear;
close all;

%% Filter Specifications
Fs = 10000;    % Sampling Frequency (Hz)
Fp = 3000;     % Passband Frequency (Hz)
Fst = 2000;    % Stopband Frequency (Hz)
Rp = 1;        % Passband Ripple (dB)
Rs = 60;       % Stopband Attenuation (dB)

%% Normalize Frequencies
Wp = Fp/(Fs/2);
Ws = Fst/(Fs/2);

%% Determine Minimum Filter Order
[N,Wn] = buttord(Wp,Ws,Rp,Rs);

%% Design Butterworth High-Pass Filter
[b,a] = butter(N,Wn,'high');

%% Display Filter Parameters
fprintf('Butterworth IIR High-Pass Filter\n');
fprintf('-----\n');
fprintf('Minimum Filter Order (N) = %d\n',N);
fprintf('Normalized Cutoff Frequency (Wn) = %.4f\n',Wn);

%% Frequency Response
figure;
freqz(b,a,1024,Fs);
```

```

title('Frequency Response');

%% Impulse Response

figure;

impz(b,a,50,Fs);

title('Impulse Response');

grid on;

%% Step Response

step_input = ones(1,50);      % Unit step signal

step_output = filter(b,a,step_input);

figure;

stem(0:49,step_output,'filled');

title('Step Response');

xlabel('Samples');

ylabel('Amplitude');

grid on;

%% Group Delay

figure;

grpdelay(b,a,1024,Fs);

title('Group Delay');

grid on;

%% Pole-Zero Plot

figure;

zplane(b,a);

title('Pole-Zero Plot');

grid on;

```

Output:



